MRC Biostatistics Unit

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Preprint

Code

DeepJoint: Robust Survival Modelling Under Clinical Presence Shift

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Motivation

- Observations' **times** and **types** reflect the patient's **condition** and the clinical staff's assessment of what was needed and when.
- Predictive models have either **partially leveraged** this ***clinical presence*** process for improved performance or assumed it **uninformative**.
- Although practitioners justify **ignoring** clinical presence to avoid leveraging a changeable and unpredictable process [1] in the hope of **improved robustness**, no work has explored the impact of handling clinical presence on robustness.
- Our work explores (i) how state-of-the-art methodologies behave under clinical presence shifts, and (ii) how to leverage clinical presence for improved transportability.

Methodology

- We study **in-ICU mortality** prediction in MIMIC III [2] leveraging **laboratory tests** after a 24-hour observation period.
- Motivated by the **difference in the counts of tests** (Fig. 3) performed on patients admitted on weekends and weekdays, we hypothesise that **practice** might **differ** between **weekends** and **weekdays** due to physicians and laboratory availability. This setting allows us to **evaluate** the **robustness to** such **change**.
- Following the evaluation procedure (Fig. 2) relying on the same normalised LOCF data, we compare the following approaches to handle clinical presence:
  - **Last** - DeepSurv multi-layer perceptron relying on the last observation.
  - **Count** - Concatenate the 24-hour count of each observation to the last observation.
  - **Ignore** - An LSTM leveraging the data while ignoring its irregularity.
  - **Resample** - An LSTM with regularly resampled data.
  - **GRU-D** - Data concatenated with missingness indicators as inputs to a GRU-D [3].
  - **Feature** - An LSTM leveraging missingness and time elapsed since previous observation as inputs.
  - **DeepJoint** - The proposed methodology (Fig. 1) leveraging the same input as Feature.

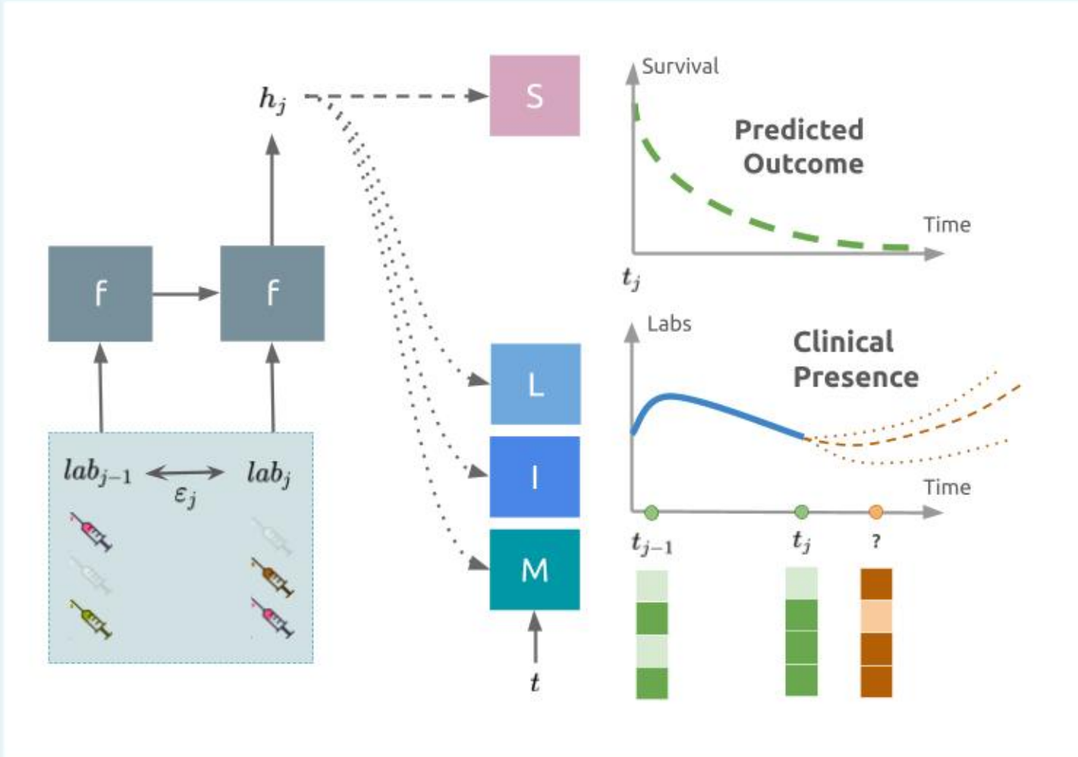

Figure 1: *DeepJoint - Joint modelling of clinical presence and survival outcome*

An LSTM network extracts an embedding ***h*** which is leveraged to model the clinical presence mechanisms through the networks ***L***, ***I***, ***M*** and the survival outcome through ***S***. The different networks aim to answer the questions:

- ➔ ***S***: *How long will the patient survive?*
- ➔ ***L***: *What is the expected next value?*
- ➔ ***I***: *When will be the next measurement?*
- ➔ ***M***: *Which laboratory test will be performed?*

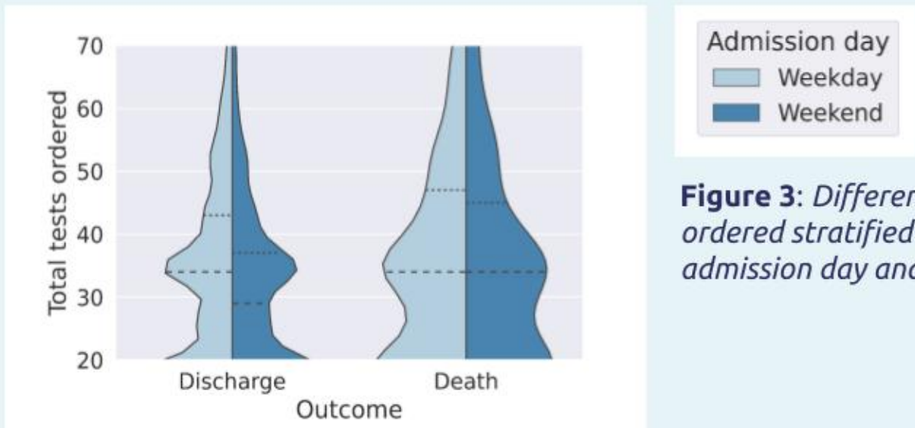

Figure 3: *Difference in tests ordered stratified by admission day and outcome*

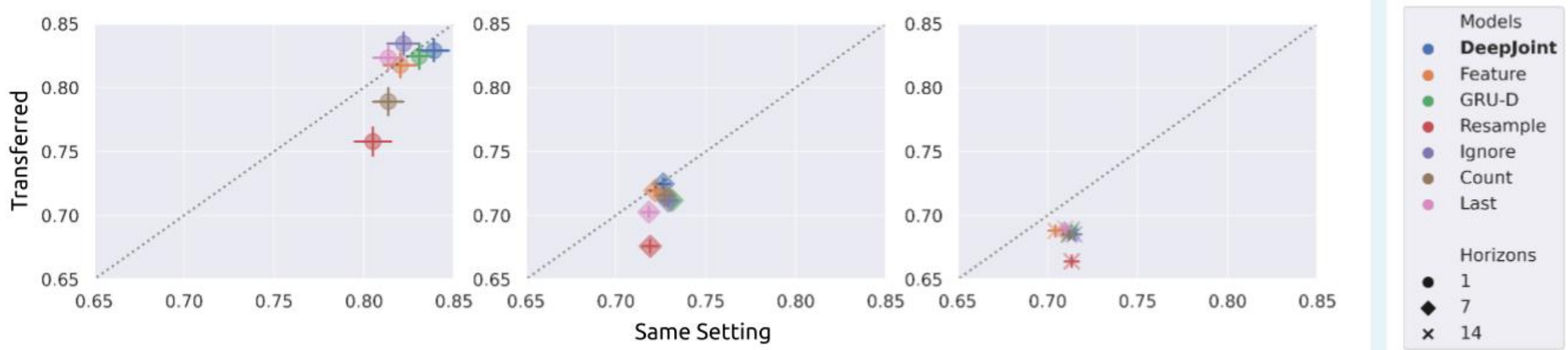

Figure 4: *Robustness performance on weekdays admissions*

One model is trained on patients admitted on weekend and another model on weekdays.
Both models are tested on the same set of weekdays patients.

Results

Figure 4 presents the transportability performance on **patients admitted on weekdays** measured through time-dependent C-index at three time horizons.

Insight 1 - Missingness and observation times are informative of short-term outcomes.

Insight 2 - Ignoring clinical presence might not only reduce performance but also hurt transportability.

Insight 3 - Explicit modelling of clinical presence improves performance and robustness.

Conclusions

Learning from medical data without sufficient **consideration** of the potential **informativeness** of **clinical presence** on **outcome** could impair performance and transportability.

We invite practitioners to:

- Explore the **practitioner-patient interaction** process to identify clinical presence patterns.
- Report the **assumptions** made on the clinical presence process.
- **Leverage** informative observation process.
- Evaluate transportability: ***internally*** and **externally**.

[1] M Van Smeden et al. *A cautionary note on the use of the missing indicator method for handling missing data in prediction research*, 2020
[2] A Johnson et al. *Mimic-III, a freely accessible critical care database*, 2016
[3] Z Che et al. *Recurrent neural networks for multivariate time series with missing values*, 2018

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